LSU | College of Art + Design

DART 7003 | Digital Culture

Brendan Harmon
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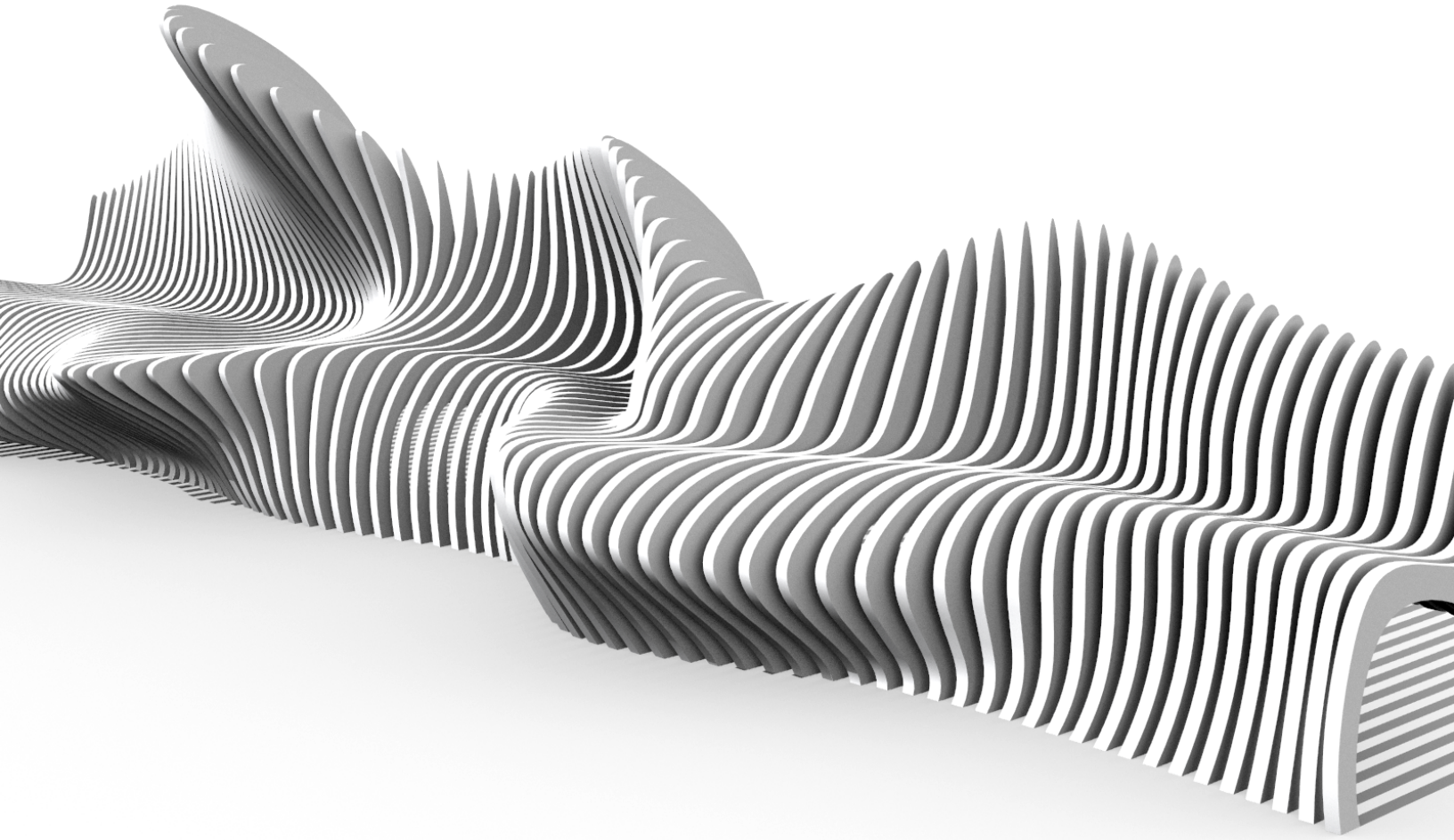
Fall 2018. Design 308.
Tuesday 6:00pm–9:00pm.



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Course Description

This course will explore how digital culture and fabrication are changing the nature of authorship and production in art and design. In seminars we will discuss cultural, technological, theoretical and ethical changes – focusing on polyvalent authorship – and in workshops you will experiment with digital methods including programming, 3D modeling, robotics, and computer numerical controlled (CNC) machining. You will put theory into practice by writing scripts that write poems and using digital fabrication to give your poems physical form. You will collect your work in an online repository with version control and present it at a exhibition in the College of Art and Design.

Topics

Open culture

1 Open source
2 Open work
3 Open science
4 Death of the Author

Digital humanities

5 Digital humanities
6 Topic modeling
7 Digital poetry
8 Sonic art

Generative design

9 Death of the architect
10 Automation & fabrication
11 Digital design
12 Robotic art

Digital fabrication

13 Digital fabrication
14 Assembly
15 Exhibition

Course Schedule

Open culture	Poetry Repository		
08.21.2018	**Open source +**	*Python:* Hello World	
08.28.2018	**Open work +**	*Python:* lists & loops	
09.04.2018	**Open science +**	*Python:* flow control	
09.11.2018	**Death of the author +**	*Python:* Markov chains	
Humanities	Poetry Generator		
09.18.2018	**Digital humanities +**	Tools & resources	Lauren Coats
09.25.2018	**Natural language processing +**	*Python:* topic modeling	
10.02.2018	**Digital poetry +**	Poetry charrette	
10.09.2018	**Sonic art +**	Poetry recordings	
Art & Design	Poetry Model		
10.16.2018	**Death of the architect +**	*Rhino:* waveforms	
10.23.2018	**Automation +**	*Rhino:* freeform modeling	
10.30.2018	**Digital design +**	*Rhino:* parametric modeling	
11.06.2018	**Robotic Art +**	Arduino	Hye Yeon Nam
Fabrication	Poetry Fabrication		
11.13.2018	**Digital fabrication**		
11.20.2018	**Assembly**		
11.27.2018	**Exhibition**		

Projects

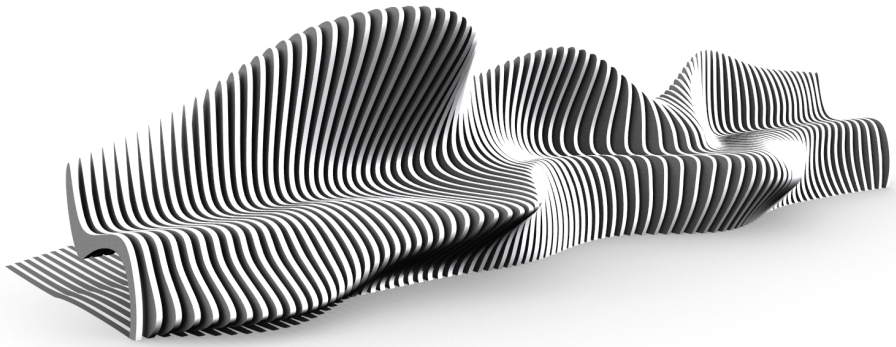
Through this series of projects explore how digital culture and fabrication are changing the nature of authorship and production.

Poetry Repository Develop a GitHub repository with a Markdown manifesto or artist's statement, Python scripts for generating poetry, Rhino 3D model, 3D renderings, and g-code for CNC milling.

Poetry Generator Develop a Python script that generates poetry. Have fun and express yourself as a meta-author!

Poetry Model Capture a series of waveforms from your poems, generate a 3D model from the waveforms, and cut the model into slices for digital fabrication. Laser-cut the slices to build a small prototype of your physical poem.

Poetry Fabrication CNC mill the slices out of sheets of birch plywood and then assemble and bond the slices to build your physical poem. Setup an exhibition showcasing your algorithmic poetry.



Grading

Poetry Repository	25%	Poetry Generator	25%
Poetry Model	25%	Poetry Fabrication	25%

Readings

Healy, Kieran. 2018. "The Plain Person's Guide to Plain Text Social Science."

Mallarmé, Stéphane. 2015. *A Roll of the Dice*. Trans. by Jeff Clark and Robert Bononno. Wave.

Ratti, Carlo, and Matthew Claudel. 2015. *Open Source Architecture*. Thames & Hudson London.

Tennant, Jon, and Ross Mounce. 2015. *Open Research Glossary*. doi:10.6084/m9.figshare.1482094.v1.

Wilf, Eitan. 2013. "Sociable Robots, Jazz Music, and Divination: Contingency as a Cultural Resource for Negotiating problems of intentionality." *American Ethnologist* 40 (4): 605–618. doi:10.1111/amet.12041.

Bohnacker, Hartmut, Benedikt Gross, Julia Laub, and Claudius Lazzeroni. 2012. *Generative Design: Visualize, Program, and Create with Processing*. Princeton Architectural Press.

Carpo, Mario. 2011. *The Alphabet and the Algorithm*. Cambridge, MA: MIT Press.

Picon, Antoine. 2010. *Digital Culture in Architecture: an Introduction for the Design Professions*. 224. Boston, MA: Birkhaeuser.

Latour, Bruno, et al. 2005. *Reassembling the Social: An Introduction to Actor-Network-Theory*. Oxford University Press.

Belsey, Catherine. 2002. *Poststructuralism: A Very Short Introduction*. OUP Oxford.

Deleuze, Gilles. 1993. *The Fold: Leibniz and the Baroque*. U of Minnesota Press.

Haraway, Donna. 1990. "A manifesto for cyborgs: Science, technology, and socialist feminism in the 1980s." *Feminism/postmodernism*: 190–233.

Eco, Umberto. 1989. *The Open Work*. Trans. by Anna Cancogni. Harvard University Press Cambridge, MA.

Barthes, Roland. 1978. *Image-Music-Text*. Macmillan.

Software

Python | https://www.python.org/
Anaconda | https://www.anaconda.com/
GitHub | https://desktop.github.com/
Rhinoceros | https://www.rhino3d.com/
Grasshopper | http://grasshopper3d.com/

Resources

Google Python Class | https://developers.google.com/edu/python/
GitHub Guides | https://guides.github.com/
Rhino Tutorials | https://vimeo.com/rhino
Grasshopper Primer | http://grasshopperprimer.com

Recommended

Fiction
J.G. Ballard, *Cloud Sculptors of Coral D*
William Gibson, *Neuromancer*
Tsutomu Nihei, *Blame!*

Cinema
Ridley Scott, *Blade Runner* & Giuliana Bruno, Ramble City
Mamoru Oshii, *Ghost in the Shell*
Alex Garland, *Ex Machina*

Art
Jean Tinguely, Machine à dessiner & Méta-Matic
François Delarozier, Les Machines de l'île, Nantes, France

Architecture
dECOi Architects, One Main, Boston, MA

Policies

Time Commitment Expectations LSU's general policy states that for each credit hour, you (the student) should plan to spend at least two hours working on course related activities outside of class. Since this course is for three credit hours, you should expect to spend a minimum of six hours outside of class each week working on assignments for this course. For more information see: http://catalog.lsu.edu/content.php?catoid=12&navoid=822.

LSU student code of conduct The LSU student code of conduct explains student rights, excused absences, and what is expected of student behavior. Students are expected to understand this code: http://students.lsu.edu/saa/students/code.

Disability Code The University is committed to making reasonable efforts to assist individuals with disabilities in their efforts to avail themselves of services and programs offered by the University. To this end, Louisiana State University will provide reasonable accommodations for persons with documented qualifying disabilities. If you have a disability and feel you need accommodations in this course, you must present a letter to me from Disability Services in 115 Johnston Hall, indicating the existence of a disability and the suggested accommodations.

Academic Integrity According to section 10.1 of the LSU Code of Student Conduct, "A student may be charged with Academic Misconduct" for a variety of offenses, including the following: unauthorized copying, collusion, or collaboration; "falsifying" data or citations; "assisting someone in the commission or attempted commission of an offense"; and plagiarism, which is defined in section 10.1.H as a "lack of appropriate citation, or the unacknowledged inclusion of someone else's words, structure, ideas, or data; failure to identify a source, or the submission of essentially the same work for two assignments without permission of the instructor(s)."

Plagiarism and Citation Method Plagiarism is the "lack of appropriate citation, or the unacknowledged inclusion of someone else's words, structure, ideas, or data; failure to identify a source, or the submission of essentially the same work for two assignments without permission of the instructor(s)" (Sec. 10.1.H of the LSU Code of Student Conduct). As a student at LSU, it is your responsibility to refrain from plagiarizing the academic property of another and to utilize appropriate citation method for all coursework. In this class, it is recommended that you use Chicago Style author-date citations. Ignorance of the citation method is not an excuse for academic misconduct.